

HAT-P-54b

R-1856

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-54_160303 · created 2026-08-03T08:51:22+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457450.6574 +/- 0.00061 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.197 +/- 0.011
Transit depth (Rp/Rs)^2	3.88 +/- 0.42 [%]
Orbital Inclination (inc)	89.96 +/- 0.83
Airmass coefficient 1 (a1)	1.009 +/- 0.02
Airmass coefficient 2 (a2)	-0.011 +/- 0.022
Scatter in the residuals of the lightcurve fit is	1.94 %
Best Comparison Star	#3 - [480, 321]
Optimal Aperture	2.96
Optimal Annulus	7.65
Transit Duration (day)	0.07574 +/- 0.00085

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.197 ± 0.011 vs 0.1572 (deviation 2.36σ)
Duration fit vs published	0.07574 d vs 0.075 d (deviation 0.57σ)
Mid-time O-C	-4.4 min
Depth SNR	11.7
Residual scatter	1.94 %

Light curve

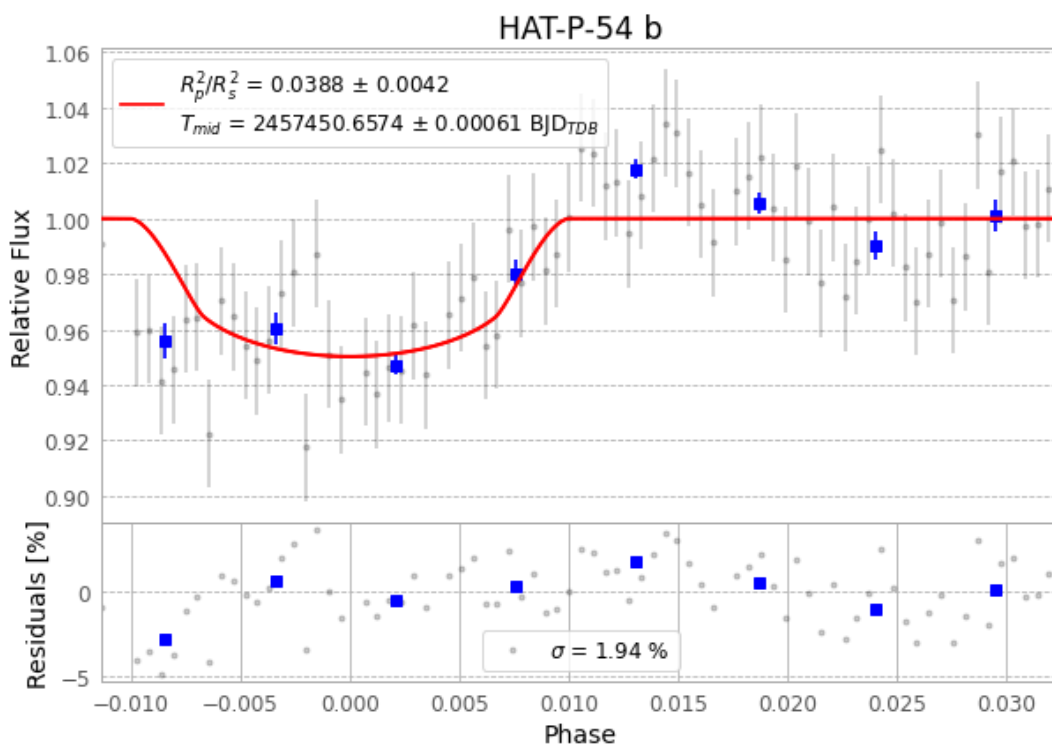


Figure 1 — FinalLightCurve_HAT-P-54 b_2016-03-02.png (SHA-256 eb7de12bfcdcbfc0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [220, 223], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 dbc9314385b7356a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

79 FITS frames (52 MB), HATP-54160303023909.FITS → HATP-54160303063912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-54-160303.log (e839b1a228cb...), FinalLightCurve_HAT-P-54 b_2016-03-02.png (eb7de12bfcdcb...), FinalLightCurve_HAT-P-54 b_2016-03-02.csv (4e96c61a64cd...)

Compute resources

Wall time 180.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 08:51Z.